Games and Information

Final Examination — Spring 2023

Prof. M. Brandt

Tuesday, May 16, 2023 · 3 hours · 100 points

Name: _____ Student ID: _____

Instructions.

- There are 5 problems worth 20 points each. Answer all of them in the blue books provided; start each problem on a new page.
- You may use two double-sided pages of handwritten notes. A basic (non-graphing) calculator is permitted.
- Justify every answer. For equilibria, state complete strategies and, where relevant, beliefs both on and off the equilibrium path.
- The exam is cumulative but emphasizes material after the midterm: auctions, signaling, and cheap talk.

Problem	1	2	3	4	5	Total
Points	20	20	20	20	20	100
Score						

Problem 1: Education as a signal (20 points)

A worker's productivity is either $\theta_H = 10$ (probability 0.4) or $\theta_L = 4$ (probability 0.6). The worker knows her productivity; firms do not. Before entering the labor market, the worker chooses a level of education $e \geq 0$, which has *no* effect on productivity. Education costs e for a low type and $e/2$ for a high type. Firms observe e (but not θ) and, because of competition, pay a wage equal to the worker's expected productivity given their beliefs: $w(e) = \mu(e) \cdot 10 + (1 - \mu(e)) \cdot 4$, where $\mu(e)$ is the belief that the worker is high-productivity. The worker's payoff is $w(e)$ minus her education cost.

(a) As a benchmark, suppose productivity were observable. What education levels and wages would we see? **[2 pts]**

(b) Find all separating PBE in pure strategies. Specify the education level of each type, the wage schedule $w(e)$ for *every* $e \geq 0$, and the beliefs that support it. **[6 pts]**

(c) Find all pooling PBE in pure strategies. **[5 pts]**

(d) Compare the payoffs of each type in the least-cost separating PBE and in the pooling PBE with $e = 0$. Which type prefers which? For what prior probability of the high type would the high type prefer the pooling PBE at $e = 0$ to the least-cost separating PBE? **[4 pts]**

(e) Show that no pooling PBE survives the Intuitive Criterion, and identify the unique separating PBE that does. **[3 pts]**

Problem 2: Three bidders (20 points)

Three risk-neutral bidders compete for a single object. Values v_1, v_2, v_3 are private and i.i.d. uniform on $[0,1]$. You may use: if $v_1, \dots, v_n$ are i.i.d. $U[0,1]$, the highest has mean $\frac{n}{n+1}$ and the second-highest has mean $\frac{n-1}{n+1}$.

(a) In a second-price auction (no reserve price), state the equilibrium bidding strategy and compute the seller's expected revenue. **[3 pts]**

(b) In a first-price auction, verify that $b(v) = \frac{2}{3}v$ is a symmetric BNE, and compute the seller's expected revenue. **[6 pts]**

(c) Consider an all-pay auction (every bidder pays her bid; the highest bid wins). Use the revenue equivalence theorem to find the symmetric equilibrium bid function and the expected revenue. State precisely which conditions of the theorem you are using. **[5 pts]**

(d) Return to the second-price auction and add a reserve price $r = \frac{1}{2}$: the object is sold only if at least one bid is at least r, and the winner pays the maximum of r and the second-highest bid. Compute the seller's expected revenue (the seller's own value is 0). Explain in one or two sentences why a reserve price can raise revenue even though it sometimes leaves the object unsold. **[6 pts]**

Problem 3: Volunteering with private costs (20 points)

A neighborhood needs one volunteer to organize a street cleanup. Each of n residents simultaneously decides whether to volunteer. If at least one resident volunteers, the cleanup happens and *every* resident gets a benefit of 1. Resident i who volunteers bears a cost c_i; volunteering by more than one resident is wasteful but allowed. Costs are private information and i.i.d. uniform on $[0,2]$. A resident who does not volunteer gets 1 if someone else volunteers and 0 otherwise.

(a) Write resident i's expected payoff from volunteering and from not volunteering, given the probability π_{-i} that at least one *other* resident volunteers. Conclude that every best response is a cutoff rule: volunteer iff $c_i \le c^*$ for some c^*. **[5 pts]**

(b) For $n = 2$, find the symmetric cutoff BNE. **[5 pts]**

(c) For $n = 3$, find the symmetric cutoff BNE. (An exact expression is required.) **[6 pts]**

(d) Compute the probability that the cleanup happens for $n = 2$ and for $n = 3$. What happens to the probability that any *given* resident volunteers as n goes from 2 to 3? Interpret. **[4 pts]**

Problem 4: Advice from a biased expert (20 points)

A state $\theta \sim U[0,1]$ is observed by an expert (sender), who sends a costless message m to a policymaker (receiver). The policymaker then chooses an action $a \in \mathbb{R}$. Payoffs are

$$u_R(a,\theta) = -(a-\theta)^2, \qquad u_S(a,\theta) = -(a-\theta-b)^2,$$

where $b = \frac{1}{10}$ is the expert's (commonly known) bias.

(a) Describe a babbling PBE and compute the policymaker's expected payoff in it. **[4 pts]**

(b) Find a PBE in which the expert reveals only whether θ is below or above a threshold x_1. Compute x_1 and the two actions induced. For which values of b does such an equilibrium exist? **[6 pts]**

(c) Compute the policymaker's expected payoff in the equilibrium of part (b) and compare it to part (a). **[4 pts]**

(d) In an equilibrium with N intervals, show that each interval is $4b$ longer than the previous one. For $b = \frac{1}{50}$, find the largest N for which an N-interval equilibrium exists and list its interval boundaries. **[6 pts]**

Problem 5: Short questions (20 points)

(a) Nature selects state X or Y with probability 1/2 each. Player 1 (rows) observes the state; Player 2 (columns) does not. Payoffs (Player 1, Player 2) are:

State X

	L	R
U	4, 2	0, 0
D	1, 0	2, 4

State Y

	L	R
U	0, 0	3, 1
D	2, 3	1, 0

Find all pure-strategy Bayesian Nash equilibria. **[6 pts]**

(b) Explain the *winner's curse* in a common-value auction. Why is it not a concern in the private-value auctions of Problem 2? **[4 pts]**

(c) Suppose the bidders in Problem 2 are risk-averse (but values are still i.i.d. private values). Which of the first-price and second-price auctions yields higher expected revenue? Give the intuition. **[4 pts]**

(d) A seller faces two bidders with i.i.d. $U[0,1]$ values and runs a second-price auction. She can either (i) set the optimal reserve price $r = \frac{1}{2}$, or (ii) attract a third bidder (same distribution) and use no reserve price. Compute the expected revenue under each option and state which is better. **[6 pts]**

— End of exam —